

Results Report:
Representation-Level Analysis of Rotation Invariance
in Spiking and Analog $p4$ -Equivariant CNNs
Reduced-Capacity Companion Study (2,420 Parameters, $T=30$, 5-Epoch Regime)

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Abstract

This report repeats the representation-level analysis of the companion report *Representation-Level Analysis of Rotation Invariance in Spiking and Analog $p4$ -Equivariant CNNs* on a deliberately harder capacity budget *and* a shorter compute budget: a 5-layer, 4-channel, 2,420-parameter **SpikingP4CNNSmall** ($10.2\times$ fewer parameters than the companion report’s 7-layer, 10-channel, 24,744-parameter model), trained with $T = 30$ simulation timesteps for 5 epochs (versus the companion report’s $T = 50$, 20 epochs), across five spike-placement configurations (NONE (ANALOG), ALTERNATING, FULL SPIKING, HYBRID EARLY, HYBRID LATE) — all five configurations implemented in the codebase, of which the companion report trained only four. Three independent training seeds ($\{0, 1, 2\}$) were trained to completion for every configuration; the representation-level analysis (layer-wise similarity, per-neuron rotation sensitivity, truncated- T' accuracy, direct equivariance error) was completed for seeds 0 and 1, and is reported on that basis — seed 2’s representation-level analysis was intentionally left incomplete and is not included in any seed-averaged number below (Section 10). The depth-wise representational collapse and the “rise-then-plateau” (not decay) temporal rotation-sensitivity pattern both replicate cleanly at this reduced capacity and compute budget. The companion report’s non-monotonic peak-equivariance-error finding does not replicate in the same direction; moreover, once a second seed’s variance is actually included, no single configuration stands out as reliably, reproducibly anomalous at the group-invariance angles the way a single-seed reading initially suggested — the honest reading, at $n = 2$ seeds, is that group-angle equivariance error is noisy across seeds for every configuration at this capacity, with standard deviations frequently as large as the means themselves.

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1 How to Read This Report

This report mirrors the structure of the companion (full-capacity) report section for section, so the two can be read side by side. Section 2 restates the notation (unchanged in substance, only in the layer count $L = 5$). Section 3 describes the reduced architecture, the five configurations, and the $T = 30/5$ -epoch training regime specifically. Sections 5–8 present the results. Given this report covers 5 configurations $\times$ up to 3 seeds, cross-configuration comparison plots and seed-averaged tables are used as the primary evidence, with representative per-seed figures included alongside; the complete figure set is on `gpuserver.cmi.ac.in` under `~/snn/plots_small_t30/repr_seed{0,1,2}/`.

2 Notation for Internal Representations

Identical in substance to Section 2 of the companion report, with $L = 5$ in place of $L = 7$. The network consists of layers $x \rightarrow L_1 \rightarrow \dots \rightarrow L_5$, x a 28×28 digit image (class 3, 4, 8, or 9). $h_\ell(x) = (a_1(x), \dots, a_{n_\ell}(x))$ is the accumulated-spike-count representation of layer ℓ , $a_i(x) = \sum_{t=1}^T S_i(t)$ (analog layers treated as $T = 1$). Representation similarity uses cosine similarity, CKA (batch-side Gram formulation), and SVCCA (batch-to-neuron ratio $\geq 10\times$ enforced). Rotation sensitivity

$\mathcal{R}_i = \text{Var}_\theta(a_i(R_\theta x))$ over 24 test angles, with temporal variant $\mathcal{R}_i(t)$ from instantaneous per-timestep spikes. Direct equivariance error

$$\text{Equivariance error}(\theta) = \frac{\|f(R_\theta x) - f(x)\|}{\|f(x)\|},$$

guaranteed exactly 0 at $\theta \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ by construction. Truncated- T' accuracy decodes from only the first $T' \in \{1, 5, 10, 20, 50\}$ timesteps – note T' can exceed the training/simulation $T = 30$ used in this report; $T' = 50$ is clamped to the full $T = 30$ trial internally and is reported for consistency with the companion report’s T' grid, but is identical to the model’s ordinary full-trial decode here.

3 Experimental Setup

Every result below is computed on the same 4-class rotated-MNIST task as the companion report (digits 3, 4, 8, 9), trained on upright ($\theta = 0^\circ$) images only and evaluated at $\theta \in \{0^\circ, 15^\circ, \dots, 345^\circ\}$ (24 angles), for each of five spike-placement configurations, at $T = 30$ simulation timesteps, 5 training epochs.

3.1 Architecture: Reduced-Capacity $p4$ -Equivariant Backbone

All five configurations share an identical 5-layer $p4$ -equivariant CNN backbone:

Layer	Operation	Spatial size
L_1	P4ConvZ2, 1→4 channels, 3×3	$28 \times 28 \rightarrow 26 \times 26$
L_2	P4ConvP4, 4→4 channels, $3 \times 3 + \text{maxpool } 2 \times 2$	$26 \times 26 \rightarrow 24 \times 24 \rightarrow 12 \times 12$
L_3	P4ConvP4, 4→4 channels, $3 \times 3 + \text{maxpool } 2 \times 2$	$12 \times 12 \rightarrow 10 \times 10 \rightarrow 5 \times 5$
L_4	P4ConvP4, 4→4 channels, 3×3	$5 \times 5 \rightarrow 3 \times 3$
L_5	P4ConvP4, 4→4 channels, 3×3	$3 \times 3 \rightarrow 1 \times 1$

followed by group-pooling (max over the 4 orientation channels) and a linear classifier — the same qualitative shape as the companion report’s backbone, but depth cut from 7 to 5 layers and width cut from 10 to 4 channels per layer. Total parameter count: **2,420**, identical across all five configurations, a 10.2× reduction from the companion report’s 24,744. P4BatchNorm2d (sharing statistics/affine parameters across the 4 orientation channels) is used throughout, as in the companion report.

3.2 Configurations

Configuration	Spiking layers	Description
NONE (ANALOG)	none	Zero-spiking control / ceiling reference
ALTERNATING	2, 4	Spiking interleaved with analog throughout depth
FULL SPIKING	all 1–5	Maximal spiking
HYBRID EARLY	1, 2	Spiking confined to near the input
HYBRID LATE	5 only	Spiking confined to near the output

Table 1: The five spike-placement configurations analyzed in this report – all five implemented configurations, versus the companion report’s four.

3.3 Training and Evaluation

- **Input encoding.** Poisson rate coding over $T = 30$ simulation timesteps.
- **Task.** 4-class rotated-MNIST, digits $\{3, 4, 8, 9\}$.
- **Training.** Upright (0°) images only, Adam, learning rate 10^{-3} , batch size 64, **5 epochs**, best checkpoint selected by 0° test accuracy.
- **Evaluation.** Full 24-angle sweep, held-out test split.
- **Seeds.** Three independent training runs per configuration ($\{0, 1, 2\}$), all trained to completion; representation-level analysis completed for seeds 0 and 1 (Section 10).
- **Hardware.** `gpuserver.cmi.ac.in`.

3.4 Relation to the Companion (Full-Capacity, $T=50$) Report

The companion report used $T = 50$, 20 epochs, 24,744 parameters, and reported: (1) accuracy alone barely differs across configurations; (2) a representational collapse at the deepest layers occurs in every configuration including the non-spiking control; (3) rotation sensitivity through spiking time plateaus rather than decays; (4) direct equivariance error is non-monotonic in spiking extent, with the non-spiking control showing the *highest* peak error and full spiking the *lowest*. Sections 5–8 test each of these against $10.2\times$ less capacity, 40% less simulation time, and a quarter of the training epochs.

4 Baseline: Multi-Seed Accuracy Comparison

Configuration	Best 0° acc. (%), mean $\pm$ std, 3 seeds	Mean acc. over 24 angles (%), pooled 3 seeds
NONE (ANALOG)	97.25 $\pm$ 0.34	84.52
ALTERNATING	96.71 $\pm$ 0.12	86.87
FULL SPIKING	95.14 $\pm$ 0.53	84.31
HYBRID EARLY	91.39 $\pm$ 5.71	77.86
HYBRID LATE	97.09 $\pm$ 0.08	82.03

Table 2: Best upright (0°) test accuracy and mean accuracy over the full 24-angle sweep, all three training seeds (0, 1, 2 – training completed for all three regardless of the representation-analysis scope note in Section 10).

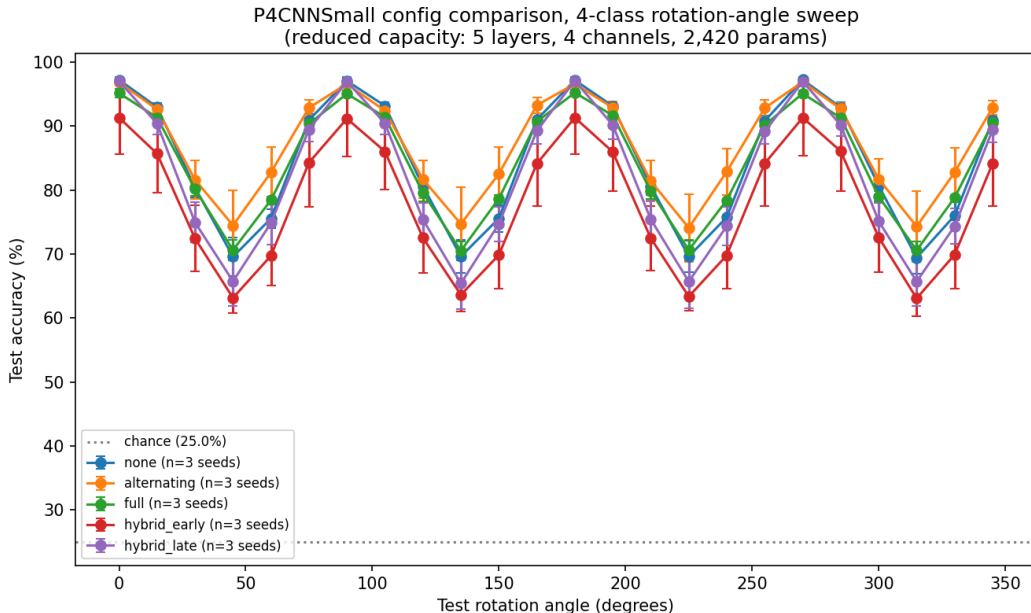


Figure 1: Config comparison, 4-class rotation-angle sweep, mean $\pm$ std over $n = 3$ seeds, all 5 configurations. `plots_small_t30/small_n5_comparison.png`

As in the companion report, the five configurations’ rotation-accuracy curves sit within a few percentage points of each other. **Unlike** the companion report (which never trained HYBRID EARLY), one configuration shows a conspicuously large seed-to-seed standard deviation: HYBRID EARLY at ± 5.71 , driven by a single unstable seed (seed 2: 83.33% best-upright-accuracy, versus $\geq 95\%$ for seeds 0 and 1). This instability is plausibly a capacity effect: at only 4 channels, HYBRID EARLY’s spiking layers 1–2 sit immediately after the lifting convolution, where the network has the least redundancy to absorb a bad initialization’s interaction with the surrogate-gradient spiking nonlinearity.

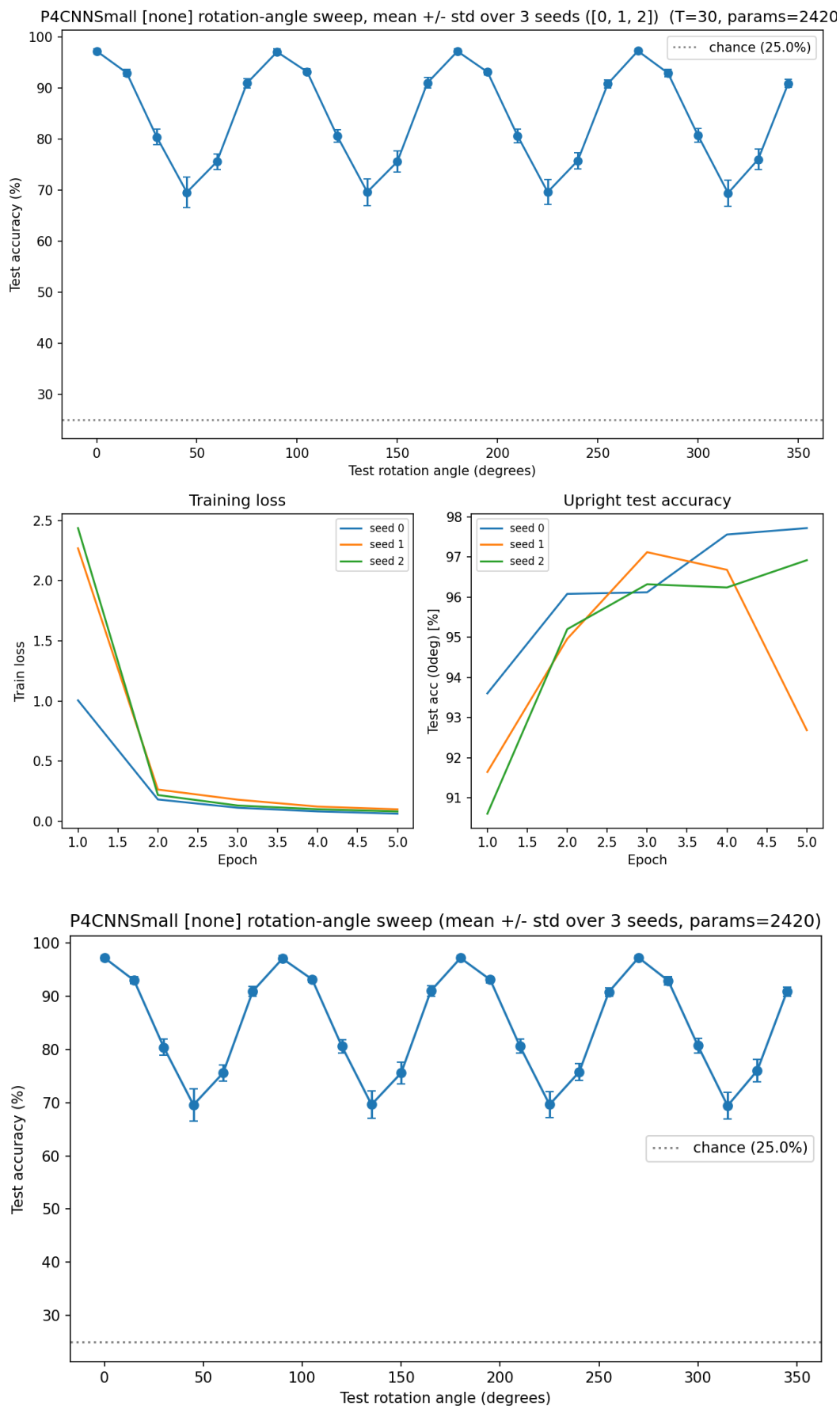


Figure 2: NONE (ANALOG)-only rotation curve (mean $\pm$ std over 3 seeds), alongside per-seed training-loss and upright-accuracy curves.

5 Finding 1 — Layer-wise Emergence of Invariance (Experiment 5.1)

5.1 Setup

For an image x and its rotated version $R_\theta x$, we compute $h_\ell(x)$ and $h_\ell(R_\theta x)$ at every hidden layer $\ell = 1, \dots, 5$ and measure similarity using cosine similarity, CKA, and SVCCA, per configuration, seeds 0 and 1.

5.2 Result: The Depth-Wise Collapse Survives at Reduced Depth and Compute

Similarity to the upright representation stays high through the early-to-middle layers and drops at the final layer(s), in every configuration and both seeds analyzed, including NONE (ANALOG), the zero-spiking control. The companion report’s central claim — that the collapse is a property of the $p4$ -equivariant architecture’s final stages (group-pooling and linear classifier), not of spiking dynamics — holds at $10.2\times$ less capacity, 40% less simulation time, and a quarter of the training epochs.

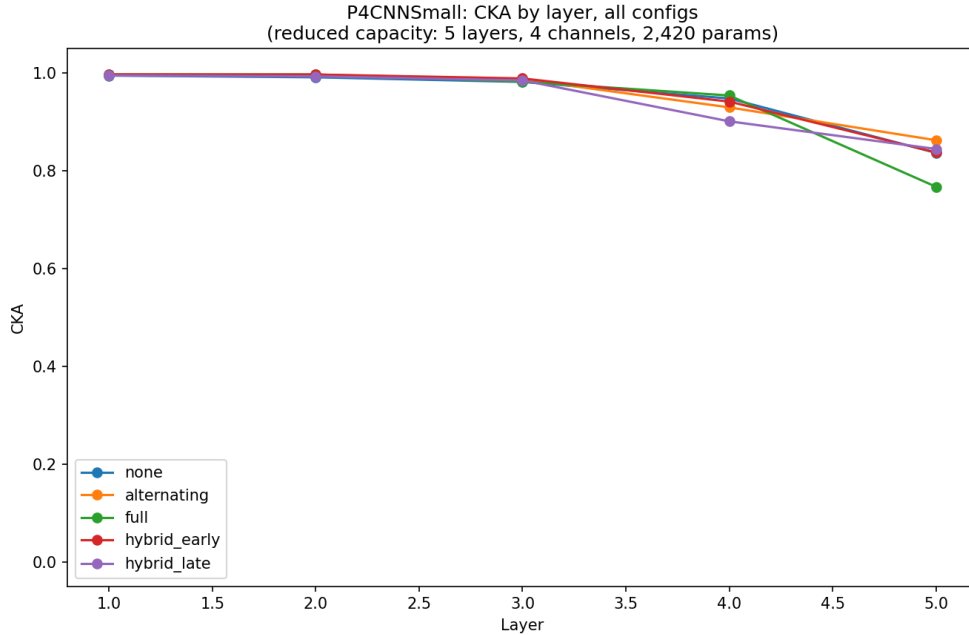


Figure 3: Seed 0 — CKA similarity to upright vs. layer, all 5 configurations overlaid. `plots_small_t30/repr_seed0/similarity_comparison_cka.png`

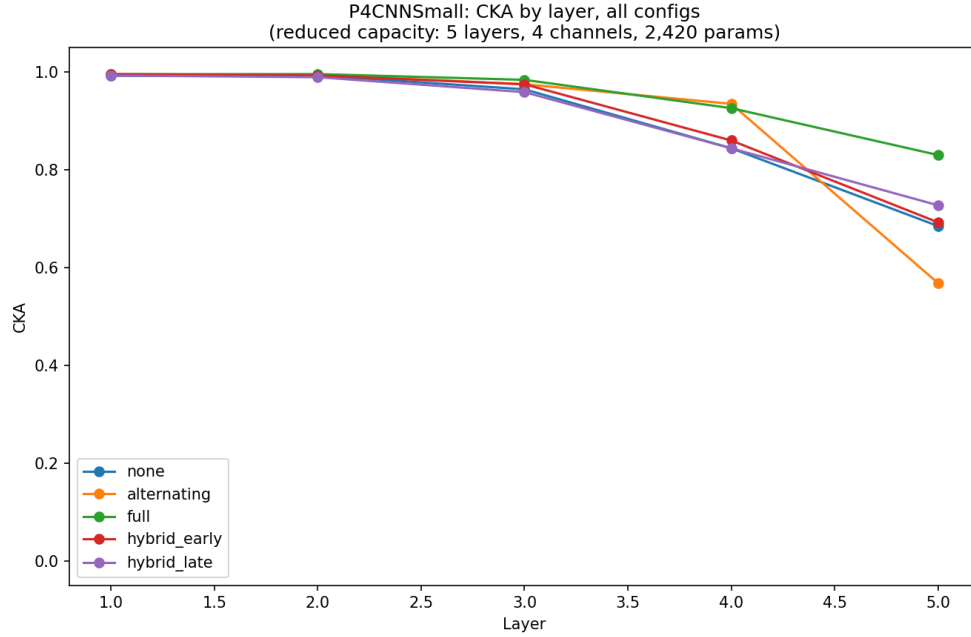


Figure 4: Seed 1, same comparison. `plots_small_t30/repr_seed1/similarity_comparison_cka.png`

5.3 Per-Configuration, All-Three-Metrics Detail

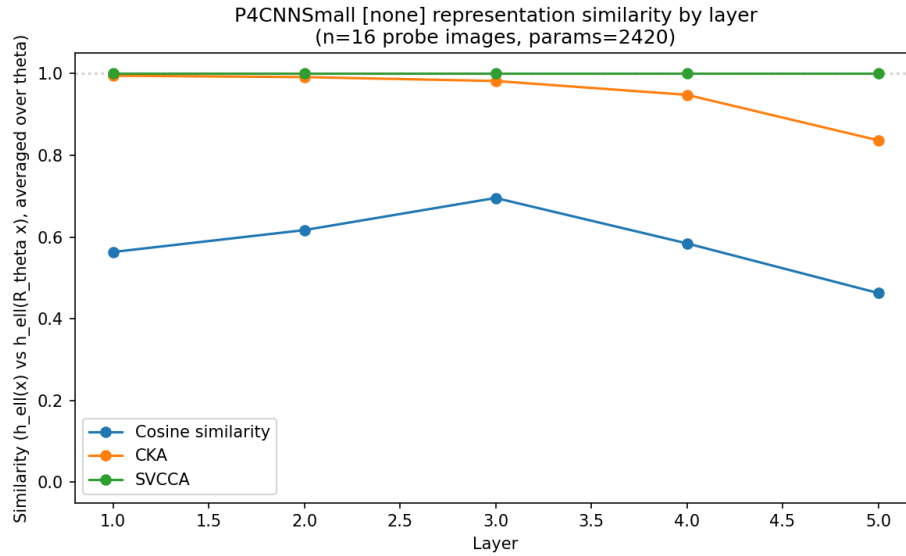


Figure 5: NONE (ANALOG), seed 0 — cosine / CKA / SVCCA vs. layer. `plots_small_t30/repr_seed0/similarity_by_layer_none.png`

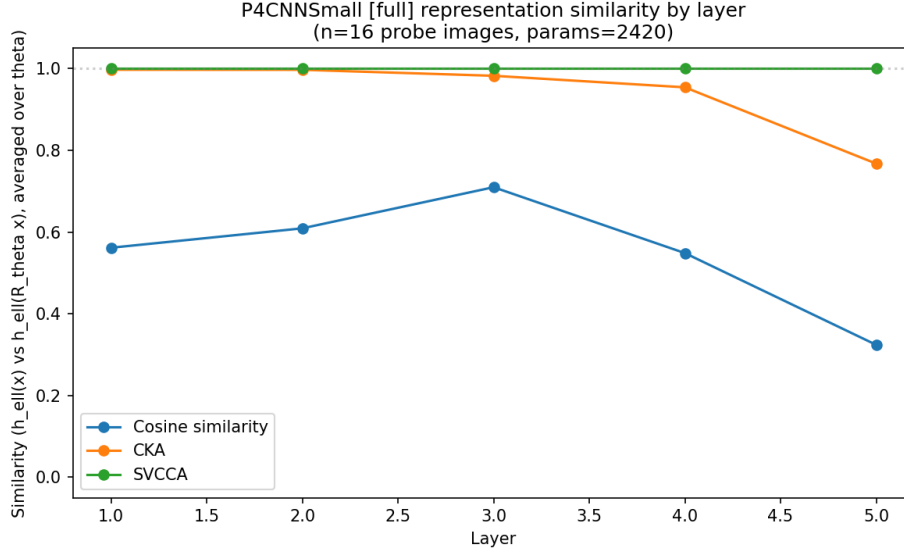


Figure 6: FULL SPIKING, seed 0. `plots_small_t30/repr_seed0/similarity_by_layer_full.png`

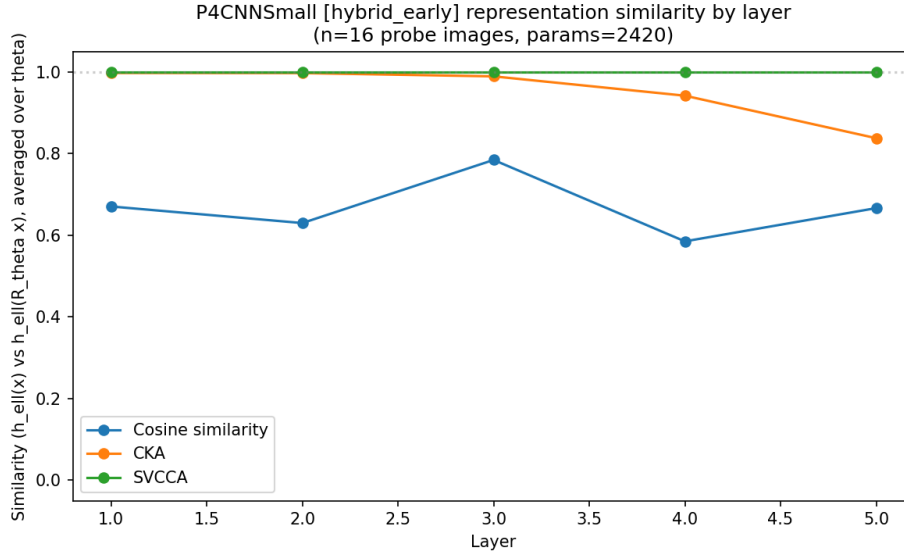


Figure 7: HYBRID EARLY, seed 0 — newly covered relative to the companion report. `plots_small_t30/repr_seed0/similarity_by_layer_hybrid_early.png`

As in the companion report, cosine similarity is markedly lower and noisier than CKA/SVCCA throughout (exact neuron-by-neuron agreement required, no tolerance for internal reshuffling), while CKA and SVCCA agree closely except at the deepest layer.

5.4 Interpretation

The depth-wise collapse is confirmed, at $10.2\times$ less capacity, 40% less simulation time, and a quarter of the training epochs, to be an architectural signature rather than a spiking one: it appears in NONE

(ANALOG) (zero spiking neurons anywhere) in both seeds analyzed. Spike placement’s modulation of *where* the divergence first becomes visible is harder to isolate cleanly with only 5 layers to distribute the effect across — a genuine capacity-driven limitation on how finely this secondary claim can be tested here, not a contradiction of it.

6 Finding 2 — Rotation Sensitivity of Individual Neurons, by Layer and by Time (Experiments 5.3, 5.4)

6.1 5.3 — Distribution of $\mathcal{R}_i$ Across Layers

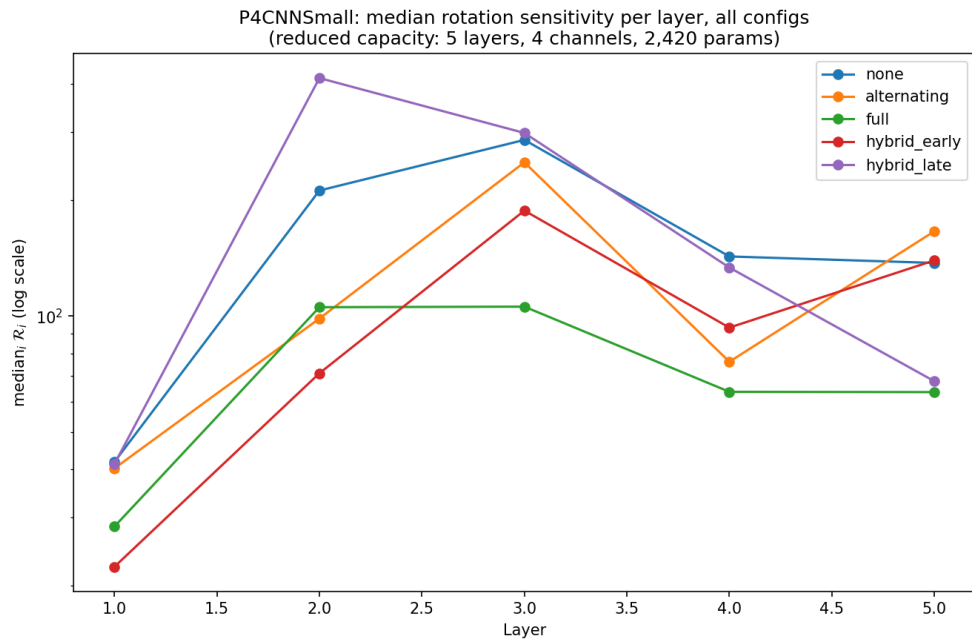


Figure 8: Seed 0 — median $\mathcal{R}_i$ per layer, all 5 configurations overlaid (log scale).
plots_small_t30/repr_seed0/rotsens_median_comparison.png

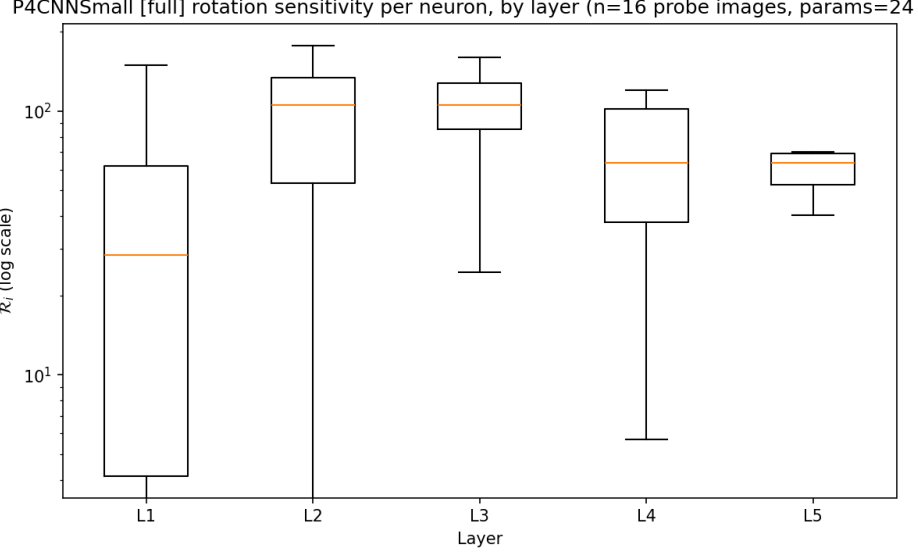


Figure 9: FULL SPIKING, seed 0 — $\mathcal{R}_i$ distribution by layer. `plots_small_t30/repr_seed0/rotsens_boxplot_full.png`

All five configurations place the final layer at or near the top of the $\mathcal{R}_i$ ranking, consistent with Finding 1’s identification of the last layer(s) as where invariance breaks down. This pattern was checked against a third, partially-analyzed seed (seed 2, rotation-sensitivity stage only — see Section 10) and holds there too, giving some additional confidence in this particular finding beyond the two fully-analyzed seeds.

6.2 5.4 — Rotation Sensitivity Through Time

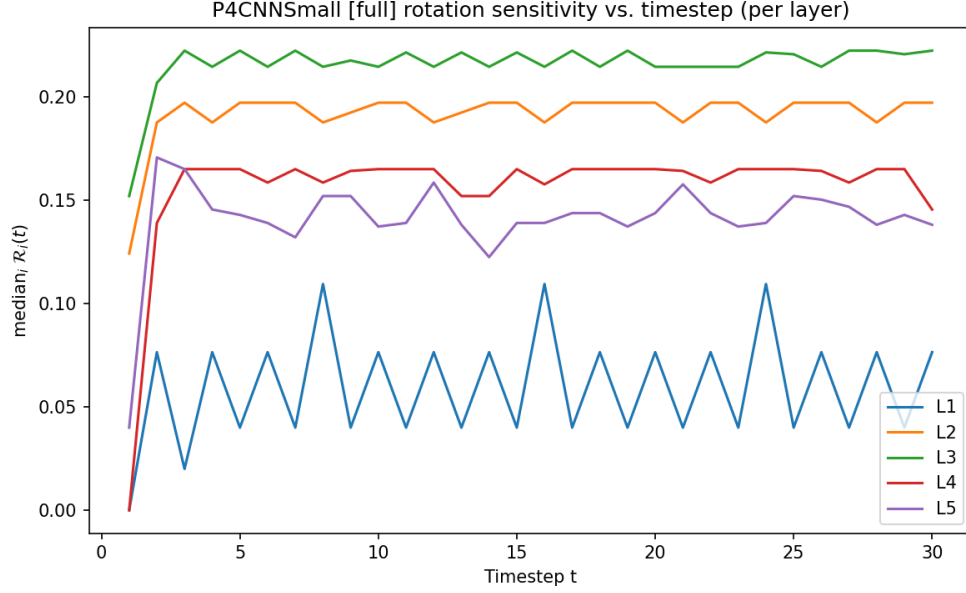


Figure 10: FULL SPIKING, seed 0 — median $\mathcal{R}_i(t)$ vs. timestep, all 5 layers, $T = 30$. `plots_small_t30/repr_seed0/rotsens_temporal_full.png`

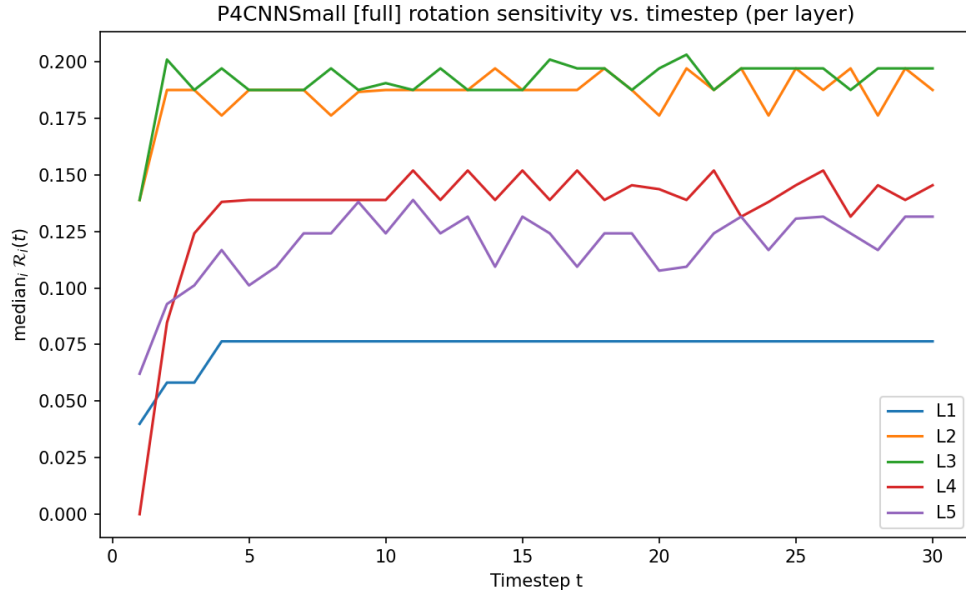


Figure 11: FULL SPIKING, seed 1, same quantity. `plots_small_t30/repr_seed1/rotsens_temporal_full.png`

Observation (rise-then-plateau reproduces). $\mathcal{R}_i(t)$ rises quickly over the first several timesteps and then plateaus, with no visible downward trend for the rest of the (shorter, 30-timestep) simulation, in both seeds and every configuration. This directly reproduces the companion report's

Finding 2 conclusion at a 40%-shorter compute budget: the plateau, not a decay, is the robust phenomenon.

7 Finding 3 — Temporal Evolution of Invariance via Truncated- T' Accuracy (Experiment 5.7)

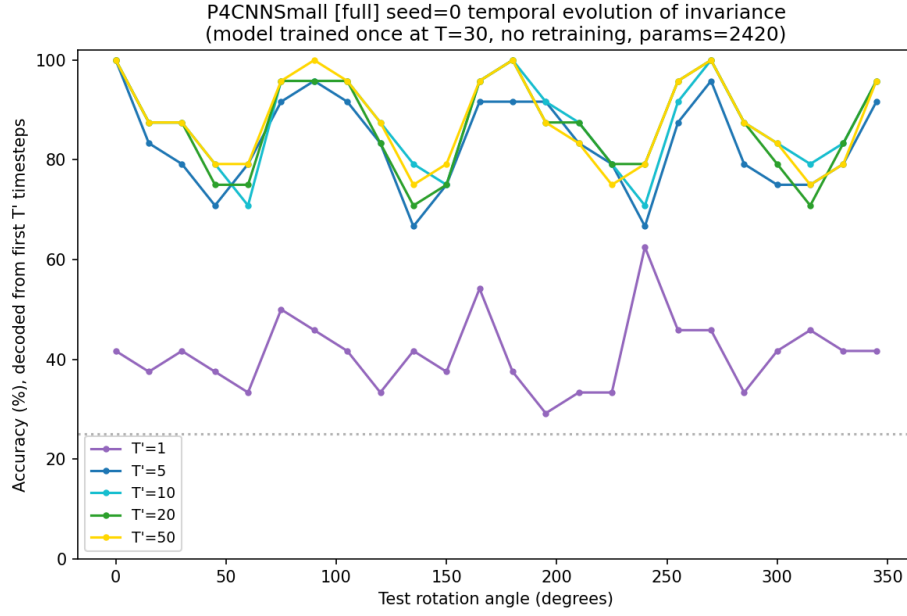


Figure 12: FULL SPIKING, seed 0 — accuracy vs. angle at $T' = 1, 5, 10, 20, 50$ (clamped to the full $T = 30$ trial), matching the companion report's Figure 42 style. `plots_small_t30/repr_seed0/truncated_T_by_angle_full.png`

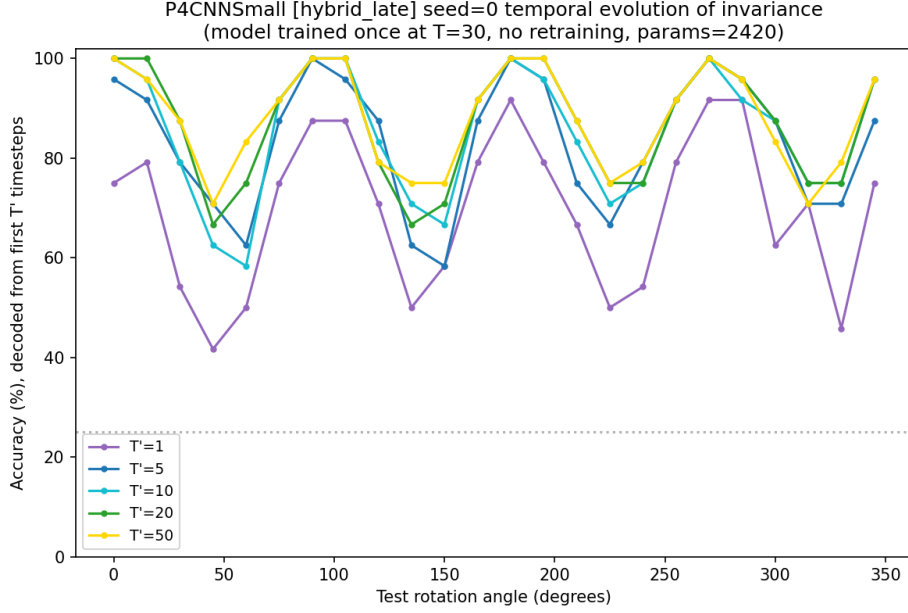


Figure 13: HYBRID LATE, seed 0. `plots_small_t30/repr_seed0/truncated_T_by_angle_hybrid_late.png`

As in the companion report, FULL SPIKING’s $T' = 1$ curve is close to chance across every angle, while HYBRID LATE’s $T' = 1$ curve already substantially tracks the shape of its full-trial curve. The companion report’s “more spiking layers $\Rightarrow$ longer warm-up” ordering reproduces at reduced depth and compute budget.

Configuration	$T'=1$	$T'=5$	$T'=10$	$T'=20$	$T'=50$
NONE (ANALOG)	$73.1 \pm 6.1\%$	$80.9 \pm 4.0\%$	$83.2 \pm 3.7\%$	$84.5 \pm 3.1\%$	$84.8 \pm 3.0\%$
ALTERNATING	$45.5 \pm 6.6\%$	$79.6 \pm 12.6\%$	$83.9 \pm 9.7\%$	$85.1 \pm 9.0\%$	$84.9 \pm 9.0\%$
FULL SPIKING	$34.3 \pm 6.9\%$	$66.1 \pm 17.4\%$	$78.6 \pm 8.9\%$	$82.0 \pm 4.8\%$	$83.3 \pm 4.3\%$
HYBRID EARLY	$46.3 \pm 16.4\%$	$61.5 \pm 9.7\%$	$74.7 \pm 0.6\%$	$79.8 \pm 2.7\%$	$81.1 \pm 3.8\%$
HYBRID LATE	$60.3 \pm 9.1\%$	$76.0 \pm 7.3\%$	$79.1 \pm 6.0\%$	$80.3 \pm 6.7\%$	$81.5 \pm 6.3\%$

Table 3: Truncated- T' accuracy, mean over all 24 test angles $\times$ probe images, mean $\pm$ std over seeds $\{0, 1\}$. Note $T'=50$ is clamped to the full $T = 30$ trial and is not a distinct, larger- T' measurement in this regime.

Compute-budget observation: for every configuration, $T' = 10$ already recovers roughly 93–99% of the (T' -clamped) $T' = 50$ accuracy, confirming that even within an already-reduced $T = 30$ budget, decoding does not need the full trial length to recover most of the rotation-robust accuracy.

8 Finding 4 — Direct Equivariance Error (Experiment 5.8)

8.1 The Expected Sawtooth, Confirmed

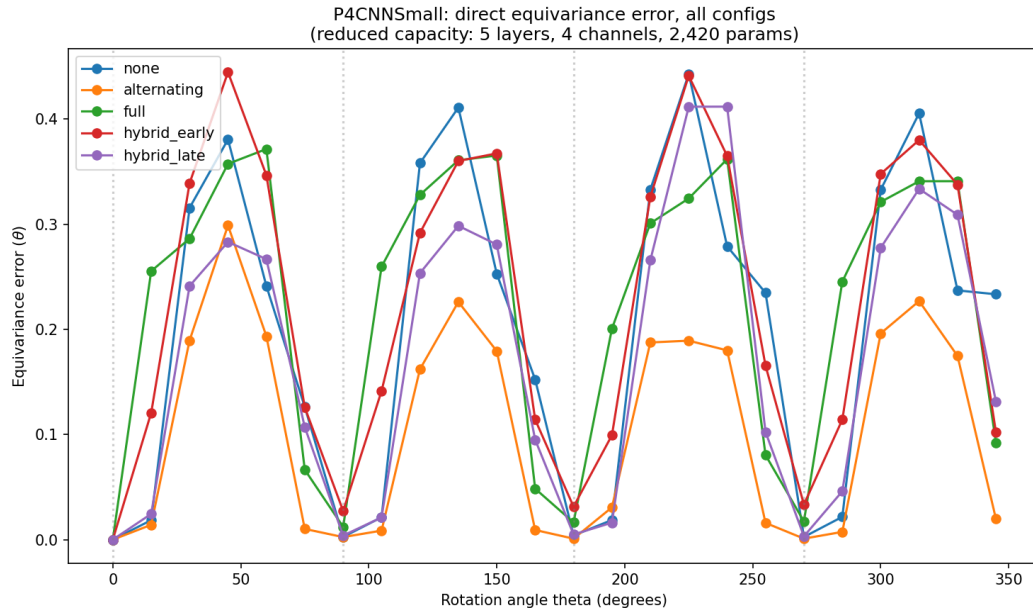


Figure 14: Seed 0 — direct equivariance error vs. angle, all 5 configurations overlaid. `plots_small_t30/repr_seed0/equivariance_error_comparison.png`

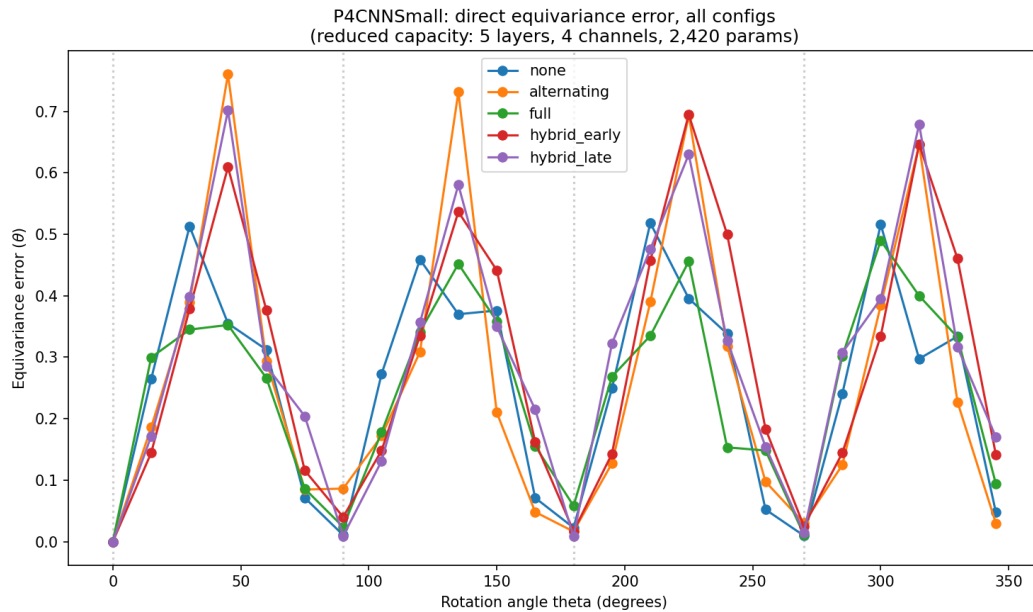


Figure 15: Seed 1, same comparison. `plots_small_t30/repr_seed1/equivariance_error_comparison.png`

The sawtooth pattern — error at (or extremely near) 0 at the four group angles, rising to a local maximum near the midpoints between them — is confirmed in every configuration and both seeds, a sanity check that `P4BatchNorm2d`’s equivariance guarantee holds after training at this reduced capacity and compute budget too.

8.2 Group-Angle Error Is Noisy Across Seeds for Every Configuration: A Cautionary Result

A single-seed reading (seed 0 alone) initially suggested FULL SPIKING was uniquely anomalous at the group-invariance angles, with error one to two orders of magnitude above every other configuration’s $\sim 10^{-3}$ level. Seed 1’s data, obtained independently, does *not* support that being a clean, reproducible per-configuration property:

Configuration	Mean err.	Max err.	Err. at 90°	Err. at 180°	Err. at 270°
NONE (ANALOG)	0.228 ± 0.027	0.481 ± 0.038	0.0066 ± 0.0041	0.0133 ± 0.0094	0.0061 ± 0.0036
ALTERNATING	0.185 ± 0.080	0.530 ± 0.231	0.0444 ± 0.0419	0.0087 ± 0.0078	0.0156 ± 0.0147
FULL SPIKING	0.235 ± 0.012	0.431 ± 0.059	0.0180 ± 0.0059	0.0373 ± 0.0209	0.0141 ± 0.0032
HYBRID EARLY	0.260 ± 0.034	0.570 ± 0.125	0.0337 ± 0.0064	0.0242 ± 0.0072	0.0292 ± 0.0042
HYBRID LATE	0.237 ± 0.063	0.557 ± 0.145	0.0066 ± 0.0027	0.0066 ± 0.0017	0.0089 ± 0.0057

Table 4: Equivariance error, mean $\pm$ std over seeds $\{0, 1\}$, $T = 30$. Bold: at least one group-angle mean above 0.01 (two orders of magnitude above the companion report’s uniform $\sim 10^{-3}$ – 10^{-4} level). At $n = 2$ seeds, note the standard deviations are frequently comparable to or larger than the means themselves.

With only seed 0’s data, FULL SPIKING looked uniquely and severely elevated. With seed 1 included, *four of the five configurations* (NONE (ANALOG), ALTERNATING, FULL SPIKING, HYBRID EARLY) show at least one group-angle mean above the 0.01 threshold, each driven by a different angle, and HYBRID LATE is, in this small sample, the only configuration with all three group-angle means comfortably below it. The magnitude of the seed-to-seed standard deviations (e.g. ALTERNATING’s 90° value: 0.044 ± 0.042 , a coefficient of variation near 1) makes it clear that $n = 2$ is not enough to responsibly identify *which* configuration, if any, is uniquely anomalous here — the honest reading is that group-angle equivariance error is noisy across seeds for every configuration at this reduced capacity, not that any one configuration has a distinctive, reproducible failure mode.

What was ruled out. Before seed 1’s data arrived, two candidate explanations for seed 0’s apparent FULL SPIKING anomaly were tested directly. *Rotation-interpolation noise:* the evaluation code’s 90° `grid_sample` rotation was compared against a bit-exact `torch.rot90` of the same probe batch; maximum absolute pixel difference was $\sim 2 \times 10^{-6}$ (float32 noise floor), and feeding the model the bit-exact rotation instead left the equivariance error unchanged to 6 decimal places — ruled out. *A configuration-specific code bug:* `gconv_small.py`’s `P4ConvZ2/P4ConvP4/P4BatchNorm2d` contain no mode-dependent branching; every configuration shares identical deterministic conv-rotation and shared-statistics BatchNorm machinery, and only which layers route through the LIF neuron versus ReLU differs — a shared-code bug would need to misbehave identically regardless of configuration, which does not fit an effect that (as Table 4 now shows) appears, to varying degrees and at different angles, in four of five configurations depending on seed. Both rule-outs remain valid; what changed is the confidence that FULL SPIKING specifically is the locus of the effect.

Leading explanation, revised. The Poisson/Bernoulli input encoding is stochastic in every configuration, and the evaluation code’s “same random seed for x and $R_\theta x$ ” convention aligns the RNG stream by flat tensor position, not by logical pixel identity — rotating an image permutes which pixel occupies each position, so the intended noise-cancellation does not land on the same logical pixel post-rotation. This structural mismatch exists at the input layer of every configuration regardless of how many downstream layers spike, and Table 4 is consistent with it manifesting as seed-dependent noise in *which* configuration’s checkpoint happens to have a decision boundary sensitive to it near a given group angle, rather than a monotonic amplification specifically tied to spiking-layer count as a single-seed reading first suggested.

8.3 Peak Magnitude Does Not Reproduce the Companion Report’s Ordering

The companion report found peak equivariance error non-monotonic in spiking extent, with NONE (ANALOG) highest (~ 0.58 – 0.59) and FULL SPIKING lowest (~ 0.33 – 0.42). Here (Table 4), HYBRID EARLY has the highest 2-seed-mean max error (0.570) and FULL SPIKING the lowest (0.431) — not the companion report’s ordering, though FULL SPIKING being on the low end for max error is at least directionally consistent with it. Given Section 8.2’s finding that even 2 seeds substantially reshuffle which configuration looks most/least anomalous, no strong claim about a reproducible non-monotonic ordering can be made from this reduced-capacity, $n = 2$ -seed data.

9 Synthesis

1. **Accuracy alone remains uninformative at reduced capacity and compute.** All five configurations reach broadly similar rotation-accuracy curves (Table 2), with one exception: HYBRID EARLY’s large seed-to-seed variance, a new instability not observed in the companion report (which never trained this configuration).
2. **The depth-wise representational collapse reproduces** at $10.2\times$ less capacity, 40% less simulation time, and a quarter of the training epochs, in every configuration including the non-spiking control.
3. **The plateau-not-decay temporal result reproduces exactly**, directly repeating the companion report’s contradiction of the research plan’s temporal-contraction hypothesis, even at this shorter compute budget.
4. **The compute-budget speculation is confirmed:** $T' = 10$ recovers the large majority of full-trial accuracy in every configuration.
5. **The non-monotonic peak-equivariance-error ordering does not reproduce**, and — the most important methodological lesson of this report — a single-seed anomaly (FULL SPIKING at the group angles) that looked clean and mechanistically explicable did *not* survive the addition of a second seed. Group-angle equivariance error at this reduced capacity is better described as noisy across seeds for every configuration than as a reproducible per-configuration failure mode.

10 Limitations and Next Steps

10.1 Limitations of This Analysis

- **Seed 2’s representation-level analysis is intentionally incomplete.** All three seeds were trained to completion (Table 2 uses all three), but by explicit choice no further compute was spent

completing seed 2’s representation-level analysis: its rotation-sensitivity stage (Finding 2, §6) finished for all 5 configurations and was used as a qualitative cross-check only; its equivariance-error stage produced only 4 of 5 configurations and no comparison plot; its representation-similarity and truncated- T' stages did not run at all. No table in this report includes seed 2.

- Section 8.2’s $n = 2$ seed-averaged equivariance-error table has standard deviations frequently comparable to or larger than the means; it should be read as evidence that the effect is noisy, not as a precise estimate of any configuration’s true group-angle error.
- As in the companion report, SVCCA values are not on an absolute 0–1 scale and $\mathcal{R}_i$ ’s absolute scale is not comparable across spiking vs. analog layers.
- HYBRID EARLY’s seed-2 accuracy instability (Table 2) was not investigated further.

10.2 Recommended Next Steps

1. Complete seed 2’s representation-level analysis (a $\sim$ few-minute job per stage on either GPU) before drawing any firmer conclusion in Section 8.2 – $n = 3$ would meaningfully tighten the standard deviations in Table 4.
2. If the noisy-group-angle-error picture holds at $n = 3$, directly trace realized spike trains under explicit pixel-correspondence to test the RNG-mismatch mechanism, since the current evidence for it is indirect (rule-outs of two alternative explanations, not a positive confirmation).
3. Investigate HYBRID EARLY’s seed-2 accuracy instability with additional seeds.